

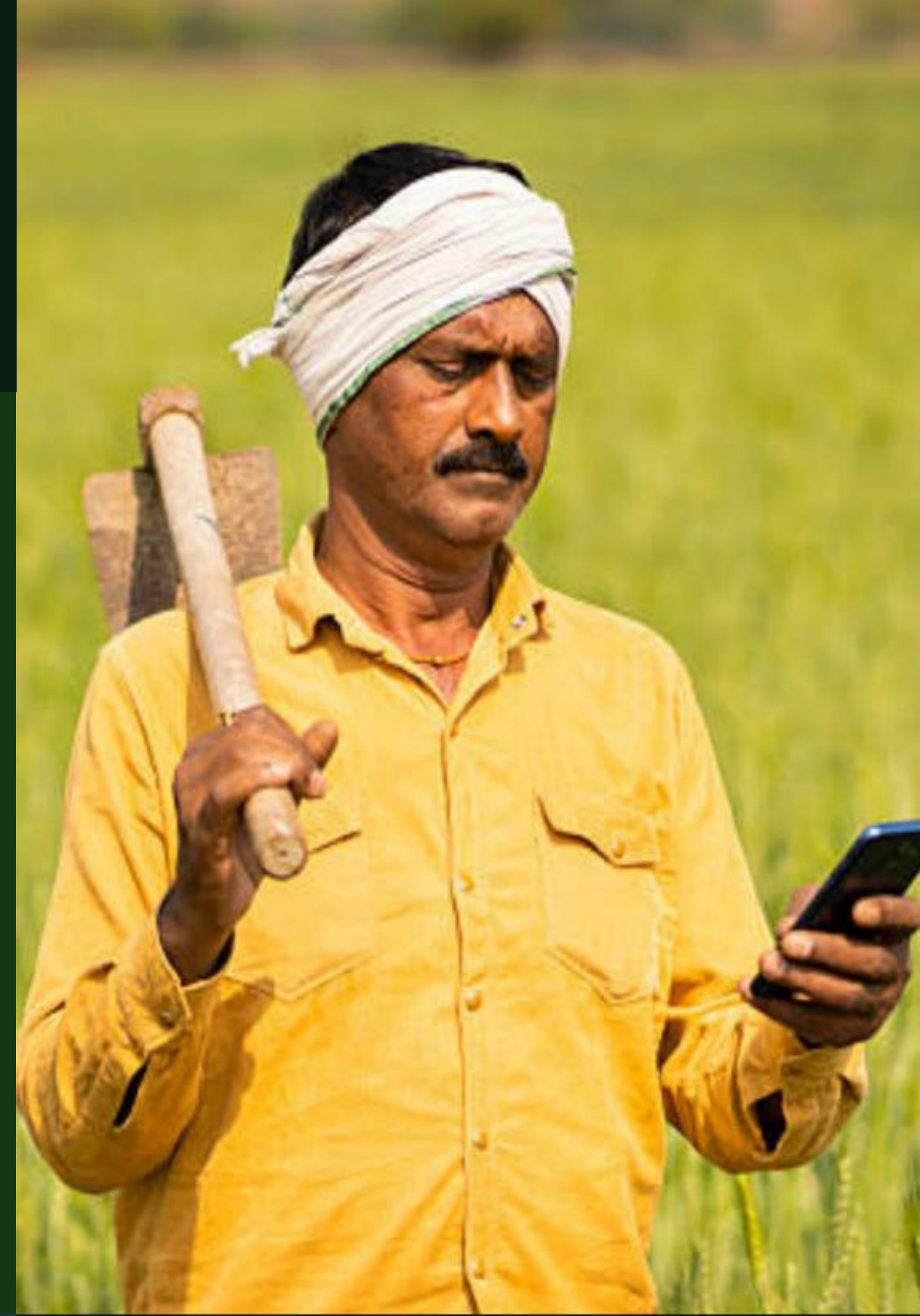
FARMDIRECT

A Direct Farmer-to-Buyer Marketplace with Hybrid Fulfilment & AI Price Intelligence

Problem Statement : Multiple intermediaries reduces farmers earnings and increases consumers prices

Team Name : NexCode

Team Members : Ritika Singh
Krishnendu Naiya



01. PROBLEM WORTH SOLVING

Fixing Supply Chain Fragmentation & High Losses



Current Fragmented Chain

Produce moves through 5-6 intermediate hand-offs before reaching end consumers, compounding costs and delays.

Farmer } Local Trader } Mandi } Wholesaler } Buyer

26.6%

Farmer share of consumer rupee
(Potato Case Study)

6 - 15%

Post-harvest loss in fruits &
vegetables (NABCONS)

FarmDirect Solution

A realistic coordination platform that eliminates avoidable hand-offs while preserving necessary logistics & aggregation.

- **Direct Market Visibility:** Transparent local demand & fair price signals.
- **Cluster Batching:** Aggregates small farmer lots into efficient delivery runs.
- **Quality Verification:** Image proof & trust ratings at pickup point

Core Design Objective

Bridge information gaps & stream line transport without forcing farmers to manage complex distribution alone.

02. TARGET AUDIENCE

Who Farm Direct Servers

Small Farmers & FPOs

Pain point: Low price transparency and dependency on local brokers.

Value: AI voice-based listing, instant price guidance, and batched pickup directly from farm gates.

(SELLERS)

Local Retailers & Kiranas

Pain point: Early morning mandi trips and unpredictable stock quality.

Value: Scheduled bulk orders with guaranteed quality specs delivered straight to their shops.

(BULK BUYERS)

Urban Households

Pain point: High store prices for aged, multi-day produce.

Value: Direct Farm Fresh delivery options with transparent origin tracking and fair pricing.

(CONSUMERS)

Cluster Logistics Partners

Pain point: Unoptimized partial loads and empty return runs.

Value: Route optimization engine providing high-density clustered pickup and drop routes.

(FULFILLMENT)

03. IDEA VISUAL SKETCH / MIROBOARD

End-to-End System Flow Architecture

MIRO BOARD CONCEPT SKETCH // FARMDIRECT FLOW



✓ Human-in-the- Loop Principle: AI suggests price & tags — Farm er confirm s/edits before publication

04. APP CAPABILITES

Core System Modules & Functions

• AI Multilingual Listing

Converts natural spoken regional audio into structured crop metadata (crop type, variety, quantity, harvest window). Eliminates manual typing barriers for rural farmers.

• AI Price Guidance Engine

Analyzes nearby mandi rates, e-NAM benchmarks, and active local platform demand to recommend fair minimum/maximum price bounds to sellers.

• Cluster Logistics & Batching

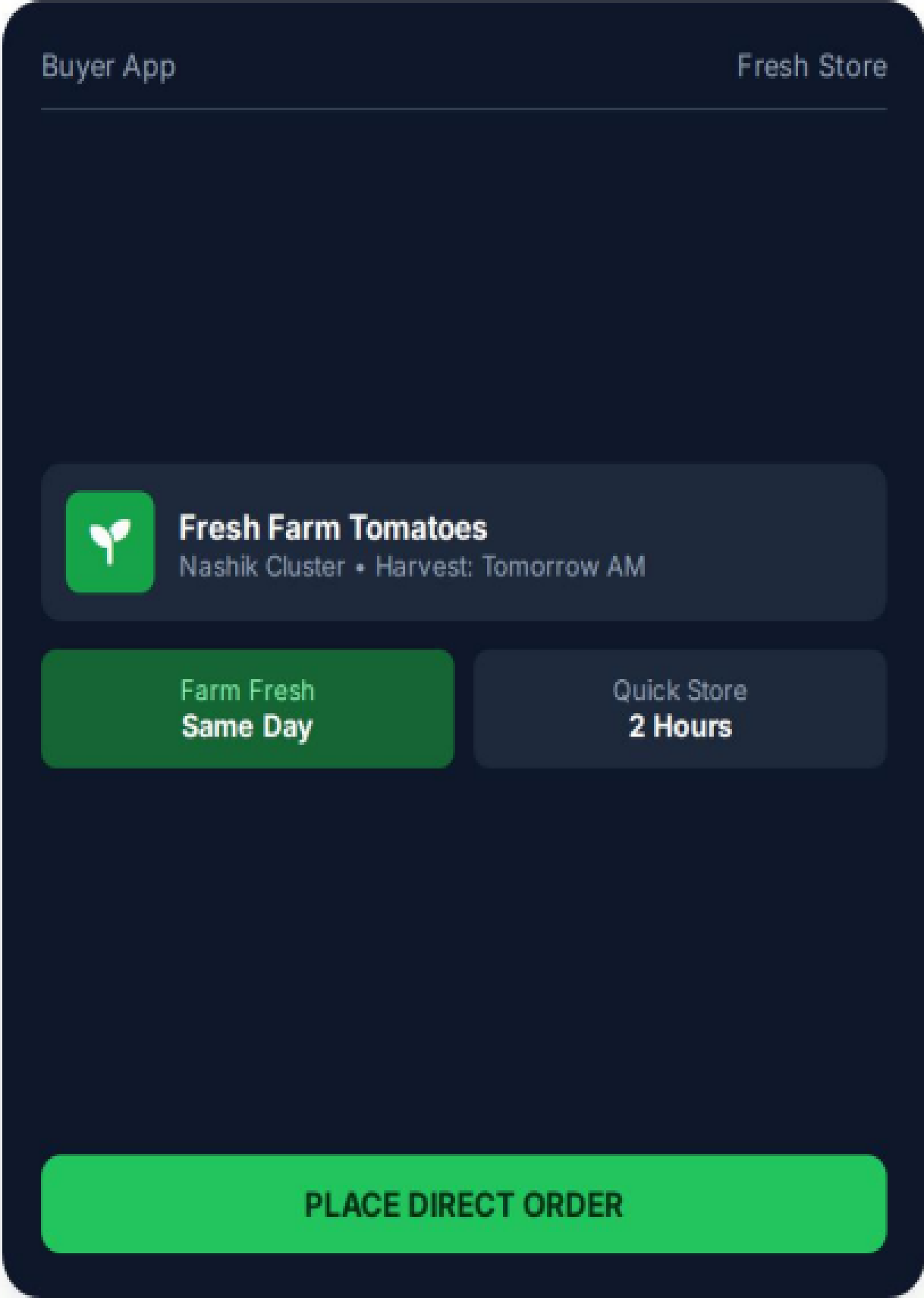
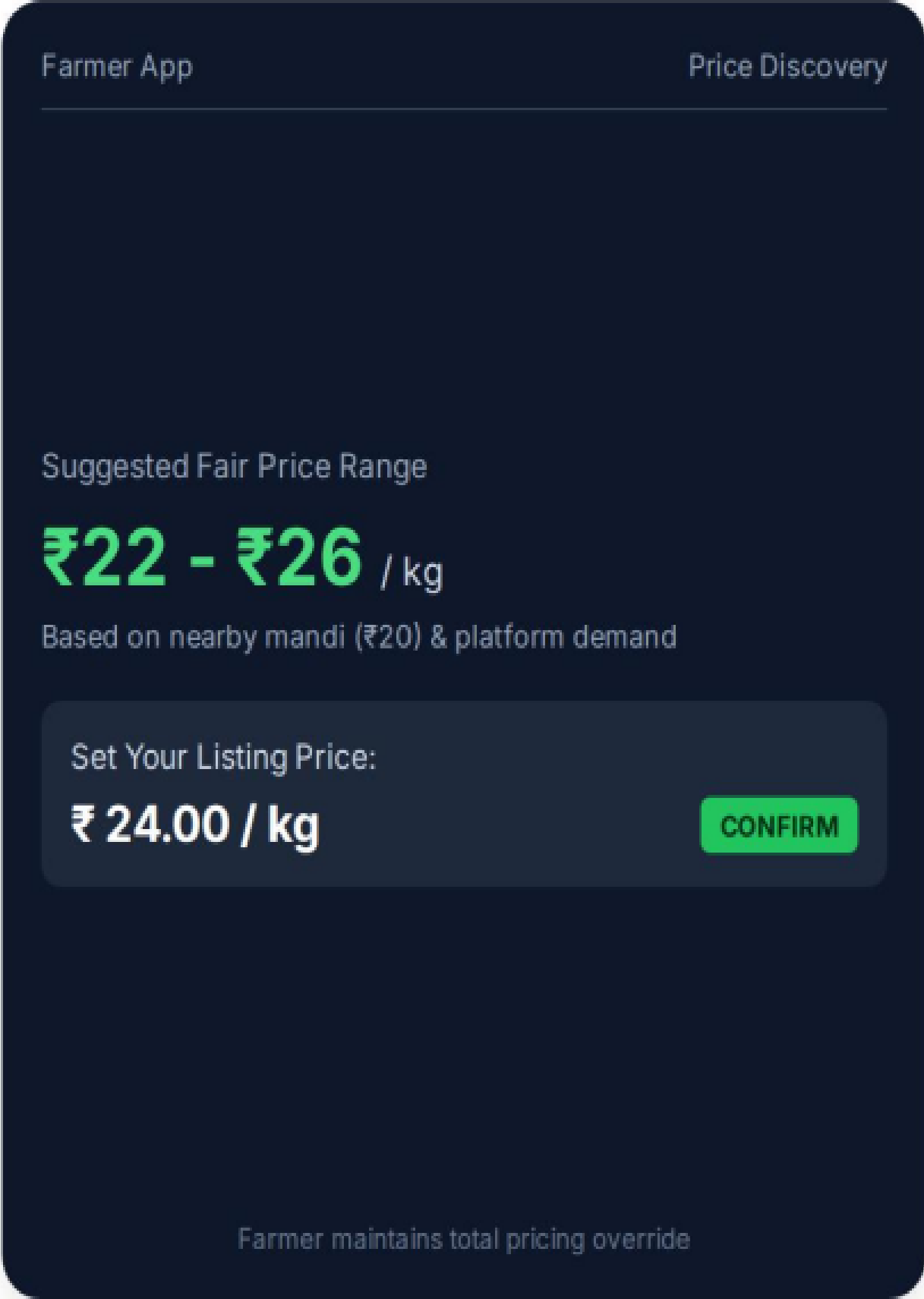
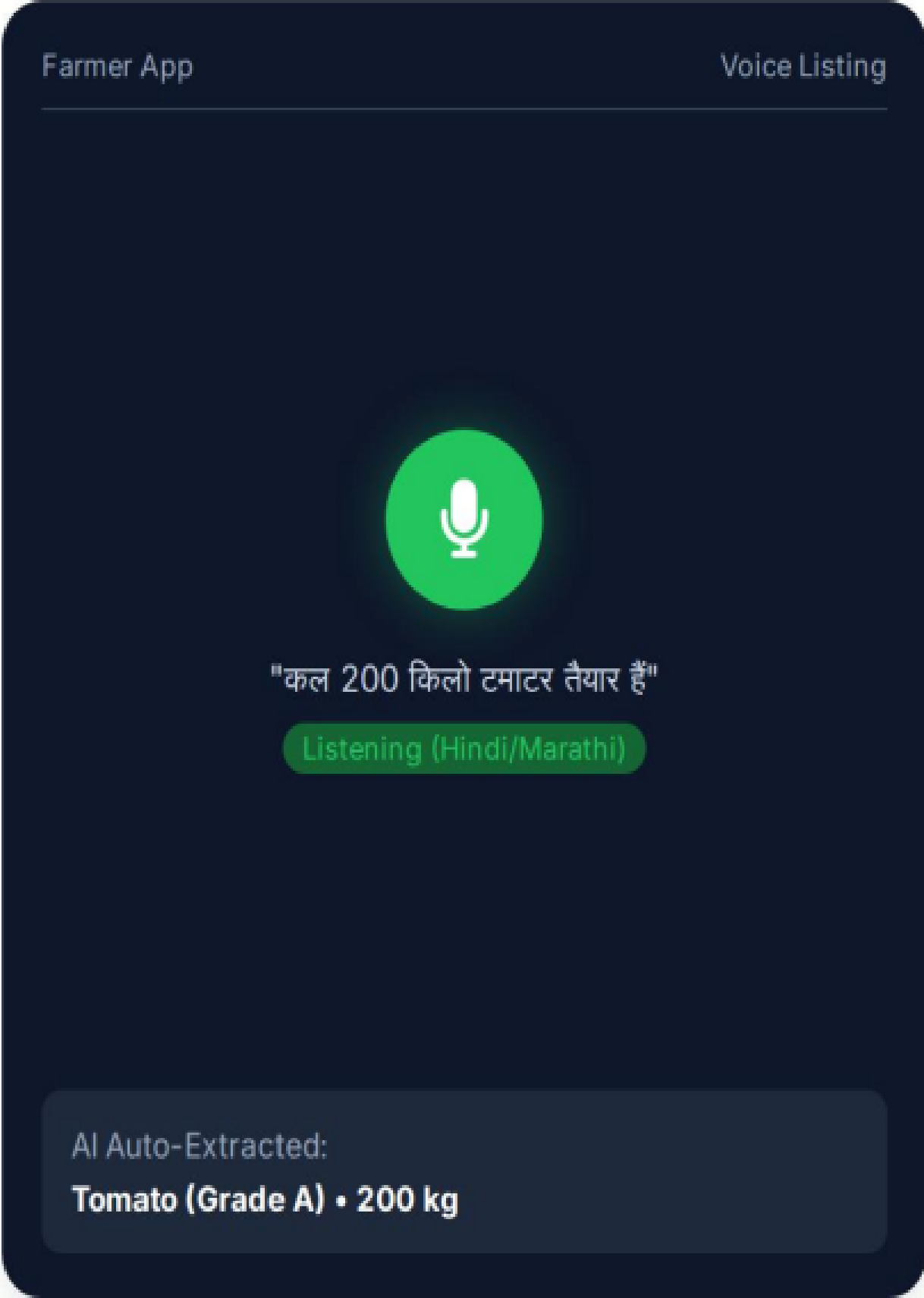
Groups neighboring smallholder orders into consolidated vehicle routes, minimizing transit costs per kilogram and reducing transport delays.

• Hybrid Fulfilment Engine

Supports dual delivery tracks: direct Farm Fresh (harvest post-order) for high freshness, and Quick Store (pre-positioned near high-density hubs) for fast turnarounds.

05. MAIN SCREENS & USER FLOW

Key Mobile UI Screen Flows



06. FULFILMENT STRATEGY

Hybrid Fulfilment: Farm Fresh vs. Quick Store

Farm Fresh Track

- Harvest and pickup occur *after* buyer order placement for maximum freshness.
- **Zero Platform Inventory Risk** : Produce stays on farm until sold.
 - **Best For** : Bulky produce, recurring household orders, and B2B buyers.
 - **Timeline** : Same-day or next-day morning dispatch.

Quick Store Track

- High-turn over fast-moving inventory pre-positioned near dense buyer clusters.
- **Demand-Driven Caps** : Pre-positioning strictly restricted by AI demand forecasts.
 - **Best For** : Urgent household daily staples (onions, potatoes, tomatoes).
 - **Timeline** : Rapid 1-2 hour neighborhood delivery.

07. TEAM STRUCTURE

Project Execution Team: Duo Structure



Full-Stack & AI Lead

Core Engineering

Mobile & Web App: Flutter / React Native cross-platform UI.
AI Integration: Whisper NLP pipeline & price guidance REST APTs.
Backend Services: Fast-API micro services & Postgre SQL database



Product & Logistics Lead

Operations & Architecture

Route Optimization: OR-Tools route batching algorithms.
User Experience: Wireframing, FPO partner onboarding, user flows.
Pilot Metrics: Benchmark evaluation & post-harvest loss tracking

08. TECHNOLOGY STACK

Modular & Scalable Tech Stack



Frontend App

React Native / Expo

Cross-platform iOS/Android support with low-bandwidth offline caching.



Backend API

FastAPI / Node.js

High-throughput asynchronous REST microservices & PostgreSQL.



AI & NLP Services

OpenAI Whisper / Llama

Speech-to-text processing for regional Indian languages.



Logistics Routing

Google OR-Tools

Vehicle routing problem solver with capacity & freshness constraints.